



**Cambridge Instruments
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

INSULATION TESTER · HAND-DRIVEN, BATTERY-FREE

CIE/444

Hand-cranked analog insulation tester in anti-magnetic aluminium body — no batteries required. Available in single-range configurations from 100V to 5000V.

BATTERY-FREE

Insulation Tester — Metal Body, Hand-Driven

A hand-driven generator-type insulation tester for direct-reading insulation resistance on machines, switchgear, cables and transformers — power is generated entirely by the operator, no batteries or external supply required.

- **No batteries required** — hand-cranked A.C. generator
- **Centrifugal governor** — holds output voltage constant
- **Anti-magnetic aluminium body** — die-cast, rugged
- **Cross-coil indicator** — anti-parallax mirror scale
- **3-second response** — quick direct reading

Supplied as base instrument only — carrying case, test leads, and range options available at extra cost.

Manufactured in India since 1963 — trusted by various State Electricity Boards, Indian Railways, cement and metal industries, and other sectors across India.



- V 100-5000V** Available ranges, specify at order
- 0 Battery-Free** Hand-cranked generator
- 3 3 Seconds** Response time

01 Full Specification Sheet

PARAMETER	SPECIFICATION
Body	Anti-magnetic aluminium (pressure die-cast)
Generator Type	Hand-driven A.C. generator, centrifugal governor
Scale Length	90 mm (approx.)
Accuracy	Conforms to IS: 2992-1987
Rotation Speed	160 r.p.m. or more
High Voltage Test	Min. 2000V or (rated V + 1000)V A.C. r.m.s., 1 min
Insulation Resistance Test	Better than 50 M Ω at 500V D.C.
Guard Terminal & Infinity Adjuster	Provided on instruments with insulation range 1000 M Ω and above
Response Time	3 seconds
Rated Voltages Available	100V / 250V / 500V / 1000V / 2500V / 5000V (single range per unit)
100V Range Options	0-20 · 0-50 · 0-100 M Ω
250V Range Options	0-50 · 0-100 · 0-250 M Ω
500V Range Options	0-100 · 0-200 · 0-500 · 0-1000 · 0-2000 M Ω
1000V Range Options	0-100 · 0-200 · 0-500 · 0-1000 · 0-2000 M Ω
2500V Range Options	0-1000 · 0-2000 · 0-2500 · 0-5000 · 0-10000 · 0-20000 · 50-50000 M Ω
5000V Range Options	0-5000 · 0-10000 · 0-20000 · 50-50000 M Ω
Dimensions (L×W×H)	110 × 128 × 195 mm (approx.)
Weight	2.5 kg (approx.)
Warranty	12 months against manufacturing defects

Any range apart from the standard voltage & resistance ranges above can be built to order.

02 Typical Applications

- Motors, generators, and other rotating machinery
- Insulation control and switching equipment checks
- Distribution network and cable insulation testing
- Transformer, insulator, and installation checks
- Locating shorts, defects, and insulation failures

03 Optional Accessories

- + Carrying case
- + Test leads, 1m/2m
- + H.V. test leads, 2m
- + Dual voltage/range option

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Instruments for accuracy and
reliability since 1963.